

Singular Fibers and Duality for the (1,3)-Debarre System

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Motivation

Families of curves often give rise to Lagrangian fibrations on hyperkähler manifolds via their **relative compactified Jacobians**. A classical example is the **Beauville–Mukai system** associated with a linear system of curves on a K3 surface. Its singular fibers, realized as compactified Jacobians of singular curves, were studied in detail by Sawon [?].

For curves on an abelian surface, the analogous construction behaves differently: the relative compactified Jacobian itself is NOT the hyperkähler manifold of interest. One must instead take a fiber of the **Albanese morphism**; this Albanese fiber is the total space of the **Debarre system**.

Main Question.

How can the singular fibers of this Albanese-reduced Lagrangian fibration be described explicitly?

More specifically, how does the Albanese condition interact with the compactified Jacobians of the three types of singular curves occurring in the (1, 3)-polarized case:

one node one cusp two nodes?

The Debarre System

We first recall the construction of the **Debarre system**.

Let (A, L) be a general (1, 3)-polarized abelian surface. Then

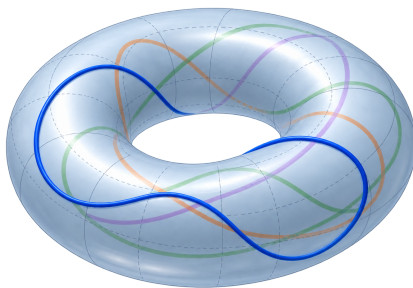
$$L^2 = 6, \quad |L| \cong \mathbb{P}^2,$$

and a smooth $C \in |L|$ has genus 4.

Let

$$\pi : \mathcal{C} \longrightarrow |L|$$

be the universal family, where the fiber over $C \in |L|$ is the curve $C \subset A$.



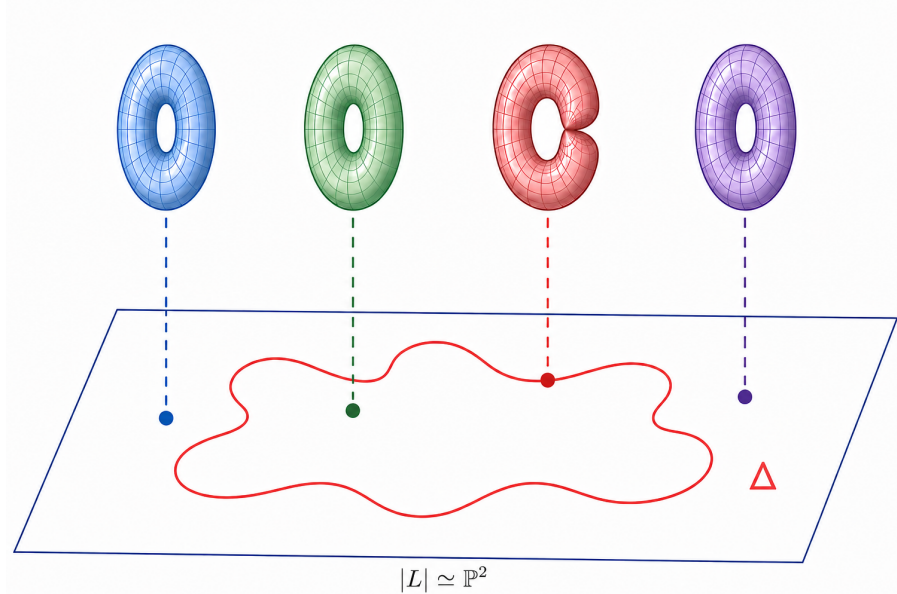
Curves $C \in |L|$ varying on a fixed abelian surface A (schematic).

For each curve C , consider its **compactified Jacobian**

$$\overline{\text{Pic}}^0(C),$$

which parametrizes rank-one torsion-free sheaves on C . These fit together into the relative compactified Jacobian

$$P := \overline{\text{Pic}}^0(\mathcal{C}/|L|).$$



Schematic picture of the relative compactified Jacobian over $|L|$, with a singular fiber over the discriminant locus Δ .

The Albanese morphism

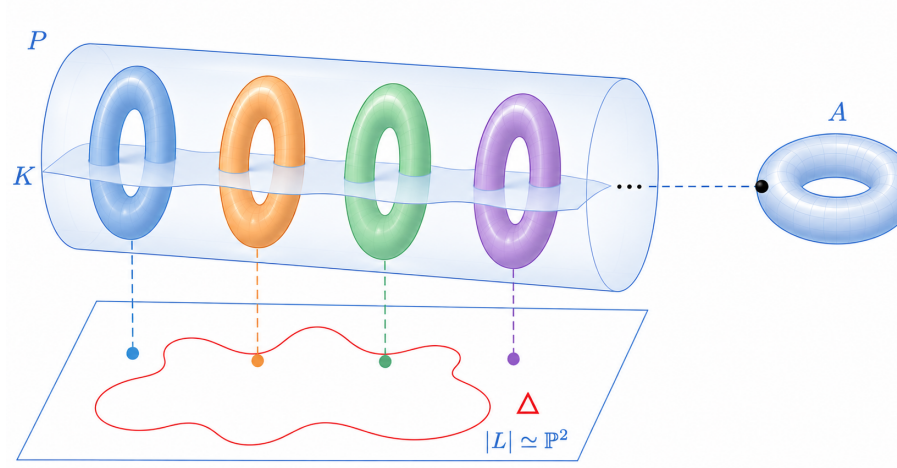
$$\alpha : P \longrightarrow A$$

is described by the Chow-theoretic second Chern character. The Debarre system is the Albanese fiber

$$K := \alpha^{-1}(0),$$

which is an **irreducible holomorphic symplectic manifold** equipped with a **Lagrangian fibration**

$$K \longrightarrow |L| \cong \mathbb{P}^2.$$



Schematic illustration of the Albanese reduction $K := \alpha^{-1}(0) \subset P$. The reduction preserves the fibration over $|L| \simeq \mathbb{P}^2$.

The Discriminant Locus

Let

$$\Delta \subset |L| \cong \mathbb{P}^2$$

be the discriminant locus parametrizing singular curves in the linear system. By Lange–Sernesi [?] and Catanese–Sernesi[?], Δ is an irreducible plane curve of degree 18 with

36 nodes and 72 cusps.

Moreover, the singularity type of a point of Δ determines the singularity type of the corresponding curve.

Point of Δ	Corresponding curve
Smooth point	One node
Cusp	One cusp
Node	Two nodes

Key observation.

Every singular fiber of the Debarre system arises from one of these three local models. Therefore, it suffices to analyze the Albanese reduction for

one node one cusp two nodes.

Compactified Jacobians of Singular Curves

For a singular curve D , the generalized Jacobian $\text{Pic}^0(D)$ is no longer proper. Allowing rank-one torsion-free sheaves gives its compactification

$$\overline{\text{Pic}}^0(D).$$

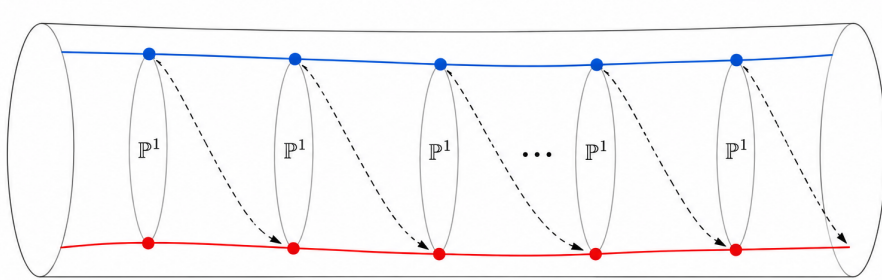
The three types of singular curves appearing in our family admit the following explicit descriptions.

One node. Let $\nu : \bar{D} \rightarrow D$ be the normalization and write $\nu^{-1}(r) = \{p, q\}$. Then

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \text{Pic}^0(D) \longrightarrow \text{Pic}^0(\bar{D}) \longrightarrow 0.$$

The normalization of $\overline{\text{Pic}}^0(D)$ is a $\mathbb{P}^1$ -bundle over $\text{Pic}^0(\bar{D})$. The two boundary sections are identified via translation by

$$\mathcal{O}_{\bar{D}}(p - q).$$

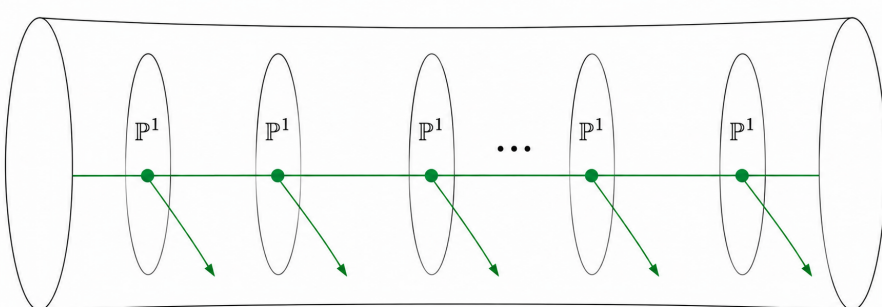


Compactified Jacobian of a one-nodal curve. It is obtained from a $\mathbb{P}^1$ -bundle over $J(\bar{C})$ by identifying the two distinguished sections via a translation.

One cusp. If D has an ordinary cusp r and $\nu^{-1}(r) = \{p\}$, then

$$0 \longrightarrow \mathbb{G}_a \longrightarrow \text{Pic}^0(D) \longrightarrow \text{Pic}^0(\bar{D}) \longrightarrow 0.$$

A smooth presentation of $\overline{\text{Pic}}^0(D)$ is given by a $\mathbb{P}^1$ -bundle associated with the first-jet bundle at p . The compactified Jacobian is obtained by contracting the distinguished vector field along the boundary section, producing a cuspidal transverse singularity.



Compactified Jacobian of a cuspidal curve. It is obtained from a $\mathbb{P}^1$ -bundle over $J(\bar{C})$ by contracting a distinguished vector field along a section.

Two nodes. Suppose that D has two nodes r_1, r_2 , with

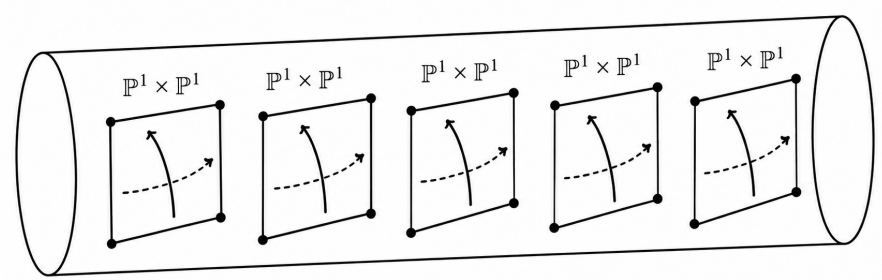
$$\nu^{-1}(r_i) = \{p_i, q_i\}, \quad i = 1, 2.$$

Then

$$0 \longrightarrow \mathbb{G}_m^2 \longrightarrow \text{Pic}^0(D) \longrightarrow \text{Pic}^0(\bar{D}) \longrightarrow 0.$$

The normalization of $\overline{\text{Pic}}^0(D)$ is a $\mathbb{P}^1 \times \mathbb{P}^1$ -bundle over $\text{Pic}^0(\bar{D})$. Its boundary divisors are identified in the two directions via translations by

$$\mathcal{O}_{\bar{D}}(p_1 - q_1), \quad \mathcal{O}_{\bar{D}}(p_2 - q_2).$$



Compactified Jacobian of a two-nodal curve. It is obtained from a $\mathbb{P}^1 \times \mathbb{P}^1$ -bundle over $J(\bar{C})$ by two independent translation gluings along the boundary sections.

The compactification is therefore controlled by

$$\mathbb{G}_m, \quad \mathbb{G}_a, \quad \mathbb{G}_m^2$$

for the node, cusp, and two-node cases, respectively.

Albanese Reduction

The relative compactified Jacobian P is larger than the Debarre system: the latter is obtained by imposing the Albanese condition. For a singular curve $D \in |L|$, the key step is to express this condition directly on $\text{Pic}^d(D)$.

Chow-theoretic Albanese map.

Let

$$i_D : D \hookrightarrow A$$

be the inclusion. A point $F \in \text{Pic}^d(D)$ is a rank-one torsion-free sheaf on D , while the Albanese morphism is defined for sheaves on the ambient abelian surface A . We therefore consider the pushforward

$$F \longmapsto (i_D)_* F.$$

Fixing a reference sheaf F_0 on D , the Albanese condition is described by the Chow-theoretic second Chern character:

$$\alpha_D(F) = \text{alb}_A \left(\text{ch}_2^{\text{CH}}((i_D)_* F) - \text{ch}_2^{\text{CH}}((i_D)_* F_0) \right).$$

Thus the passage

$$F \longmapsto (i_D)_* F \longmapsto \text{ch}_2^{\text{CH}}((i_D)_* F) \longmapsto A$$

gives an explicit description of the Albanese condition on each compactified Jacobian.

Reduction on the normalization.

Let

$$\nu : \bar{D} \longrightarrow D$$

be the normalization. Applying the above formula to sheaves obtained from line bundles on $\bar{D}$ yields the homomorphism

$$\Phi_D : \text{Pic}^0(\bar{D}) \longrightarrow A.$$

Set

$$B_D := \ker(\Phi_D)^0.$$

Since every homomorphism from $\mathbb{G}_m$ or $\mathbb{G}_a$ to an abelian variety is trivial, the Albanese map kills the affine part of the generalized Jacobian and factors through $\text{Pic}^0(\bar{D})$. Then, the Albanese condition on the presentation of $\text{Pic}^d(D)$ is therefore equivalent to restricting its base from

$$\text{Pic}^0(\bar{D}) \quad \text{to} \quad B_D.$$

Compatibility with the boundary.

Crucially, this restriction is compatible with the boundary identifications defining the compactified Jacobian. Hence the singular fiber of the Debarre system is obtained by carrying out the same compactification procedure over B_D :

node	: translation gluing,
cusp	: vector-field contraction,
two nodes	: two translation gluings.

Main Theorem

Let $D \in |L|$ be a singular curve and let

$$K_D := \alpha_D^{-1}(0)$$

be the corresponding singular fiber of the Debarre system.

Then K_D is obtained from the standard presentation of $\overline{\text{Pic}}^0(D)$ by restricting its base to the Albanese kernel and performing the same boundary identifications or contractions.

More precisely:

- **One node:** a $\mathbb{P}^1$ -bundle over the Albanese kernel, with the two boundary sections identified by the induced translation;
- **One cusp:** the corresponding $\mathbb{P}^1$ -bundle over the Albanese kernel, with the distinguished tangent direction along the boundary contracted;
- **Two nodes:** a $\mathbb{P}^1 \times \mathbb{P}^1$ -bundle over the Albanese kernel, with the two pairs of boundary divisors identified by the induced translations.

Dual Lagrangian Fibration

A distinctive feature of the Debarre system is that its smooth fibers are not principally polarized: they carry a polarization of type (1, 3). Consequently, unlike Beauville–Mukai systems, the fibration is NOT fiberwise self-dual.

Two quotient constructions.

Let $D \in |L|$ be smooth and write

$$0 \longrightarrow K_D \longrightarrow \text{Pic}^0(D) \xrightarrow{\alpha_D} A \longrightarrow 0.$$

There are two natural descriptions of the dual fiber.

Sawon [?]: $\bar{K}_D \simeq \text{Pic}^0(D)/\bar{A}$, $\bar{A} = \text{Pic}^0(A)$,
Kim [?]: $\bar{K}_D \simeq K_D/G_D$, $G_D = \ker(\lambda_D)$,

where

$$\lambda_D : K_D \longrightarrow \bar{K}_D$$

is the polarization isogeny.

For the Debarre system these two quotient constructions agree: the action of $\bar{A}$ on the ambient Jacobian moves the Albanese fibers transitively, while its stabilizer on K_D is precisely G_D . Hence

$$\text{Pic}^0(D)/\bar{A} \simeq K_D/G_D.$$

Failure of self-duality.

For a (1, 3)-polarized Debarre fiber,

$$G_D \simeq (\mathbb{Z}/3\mathbb{Z})^2.$$

Thus the dual fiber is obtained by a non-trivial finite quotient

$$\bar{K}_D \simeq K_D/G_D,$$

in sharp contrast with the principally polarized Beauville–Mukai case, where the corresponding kernel is trivial and the smooth fibers are self-dual.

Orbifold nature of the dual.

Since the polarization is of type (1, 3), the dual construction necessarily involves a non-trivial finite quotient. Thus, in contrast with the self-dual Beauville–Mukai case, one should not expect the dual Debarre system to remain smooth.

This naturally leads to an **orbifold version** of the dual Lagrangian fibration, rather than another smooth hyperkähler manifold.

Outlook

The explicit description of the singular fibers opens several directions for further study.

Monodromy. The three degenerations—node, cusp, and two nodes—provide local models for computing the monodromy around the discriminant. Understanding these local transformations is a first step toward determining the global monodromy of the Debarre system.

Geometry of the dual fibers. Since the dual fibration is a non-trivial finite quotient

$$\bar{K} \simeq K/G,$$

it is natural to study how this global finite group acts on the singular fibers. In particular, one can compare the node, cusp, and two-node fibers with their counterparts in the dual fibration.

Fourier–Mukai duality.

A further goal is to construct a relative Poincaré sheaf across the singular fibers. A global universal sheaf may be obstructed by a Brauer class

$$\alpha \in \text{Br}(\bar{K}),$$

leading naturally to an α -twisted Poincaré sheaf and a candidate for a **relative Fourier–Mukai equivalence**

$$D^b(K) \stackrel{?}{\simeq} D^b(\bar{K}, \alpha).$$

References

References

- [CS22] F. Catanese and E. Sernesi, *The Hesse pencil and polarizations of type (1, 3) on Abelian surfaces*, arXiv: 2212.14877.
- [K25] Y.-J. Kim, *The dual Lagrangian fibration of known hyper-Kähler manifolds*, Algebraic Geometry **12** (2025), no. 5, 661–700.
- [LS02] H. Lange and E. Sernesi, *Severi varieties and branch curves of abelian surfaces of type (1, 3)*, Int. J. Math. **13** (2002), no. 3, 227–244.
- [S08] J. Sawon, *Twisted Fourier–Mukai transforms for holomorphic symplectic four-folds*, Adv. Math. **218** (2008), no. 3, 828–864.
- [S20] J. Sawon, *Lagrangian fibrations by Prym varieties*, Matematică Contemporană **47** (2020), 182–227.

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